

Agentic AI in Logistics

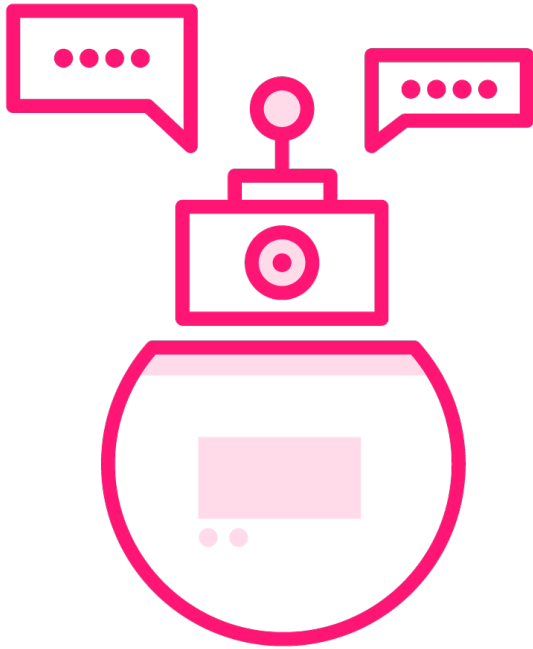


Laurentiu Raducu

Founder of bitheap

@bitheap_tech | www.bitheap.ch



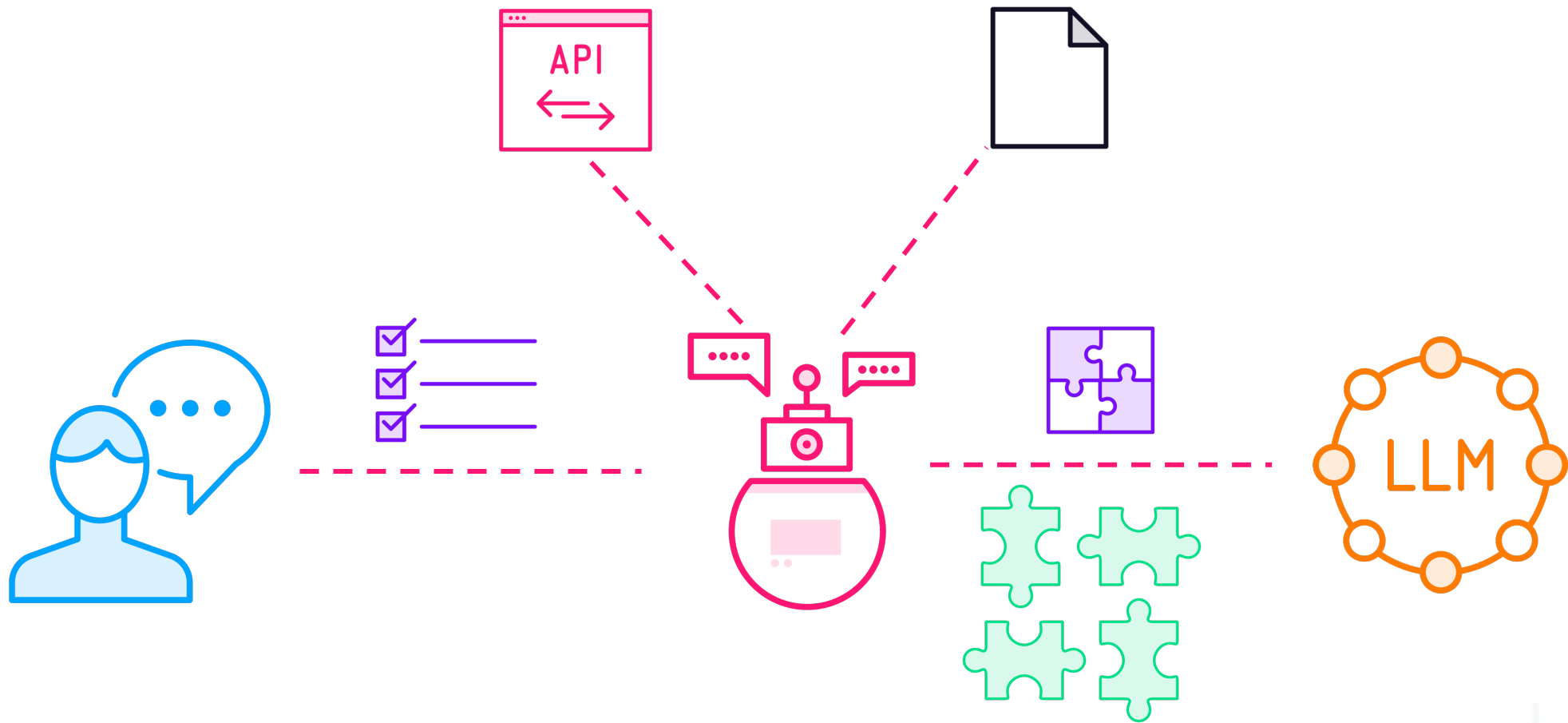


AI agent tech has huge potential

Organizations invest heavily in setting up agents

They perform tasks autonomously





AI Agents in Production

Customer support

Software development



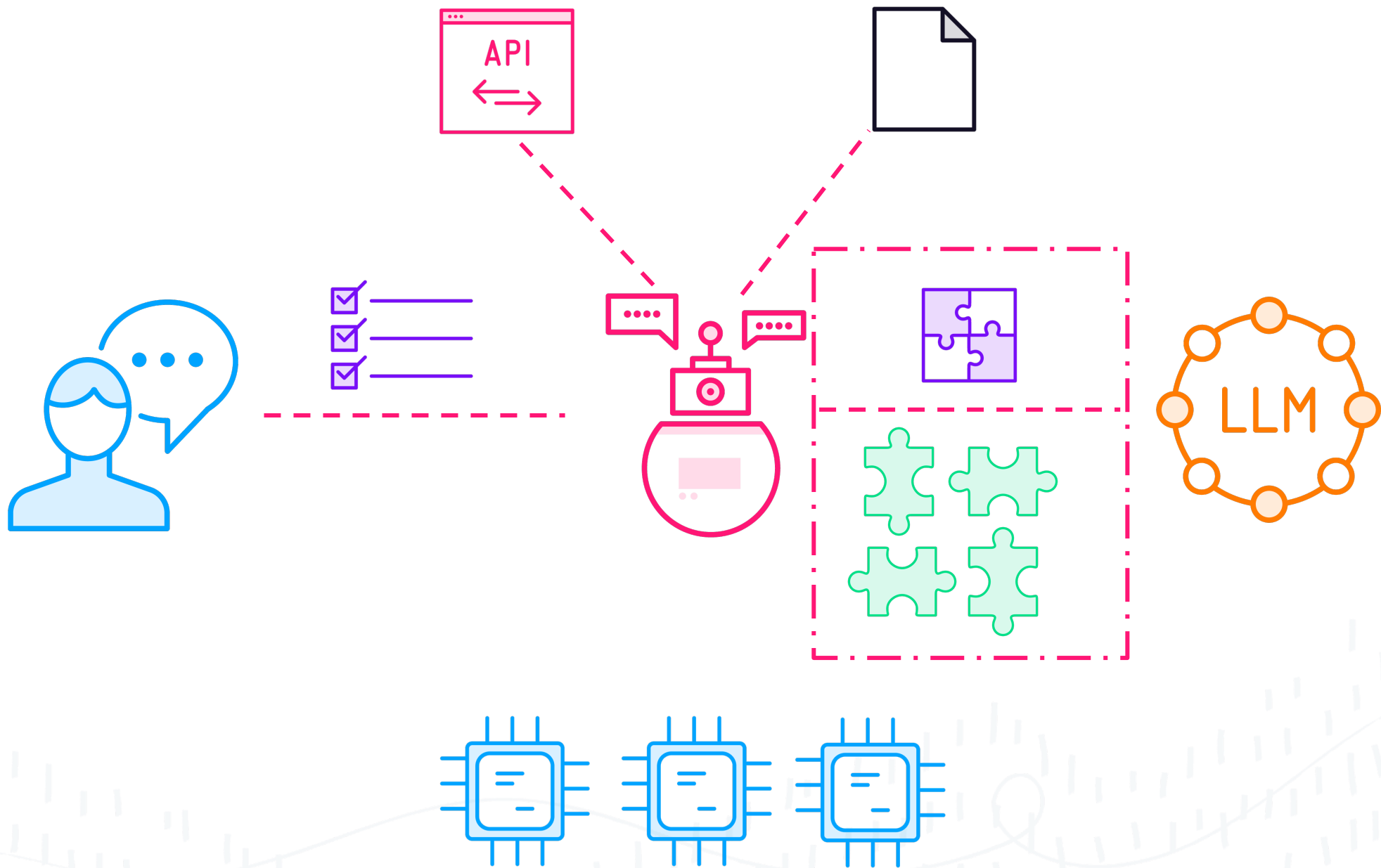
Limitations

Limited context
window

Hallucinations

Costs

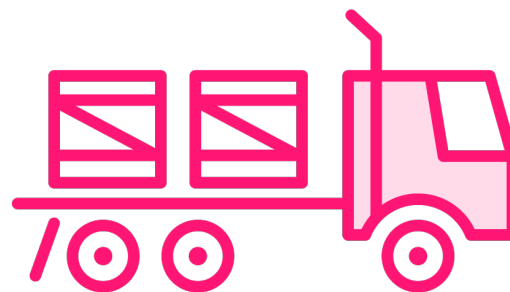
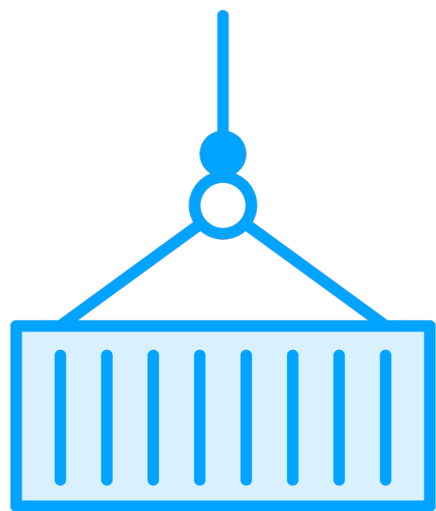






Logistics & Supply Chain Challenges





Suboptimal Transportation Planning

**Relying on static
maps and manual
processes**

**Ignore real-time
factors**

“Empty miles”

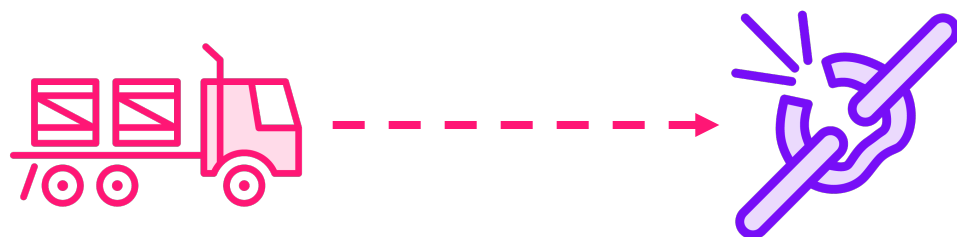


Effects

High costs and longer lead times

Reduced capacity utilization





Inventory Mismanagement

Poor forecasting

**No real-time
visibility on
inventory**

**Poor lifecycle
management**



Effects

**Significant capital is locked away
in excess stock**

**Raised warehousing, insurance
and obsolescence costs**



Poor Demand Forecasting

**Over-relying on
historical data**

**Ignoring market
shifts, promotions, or
disruptions**

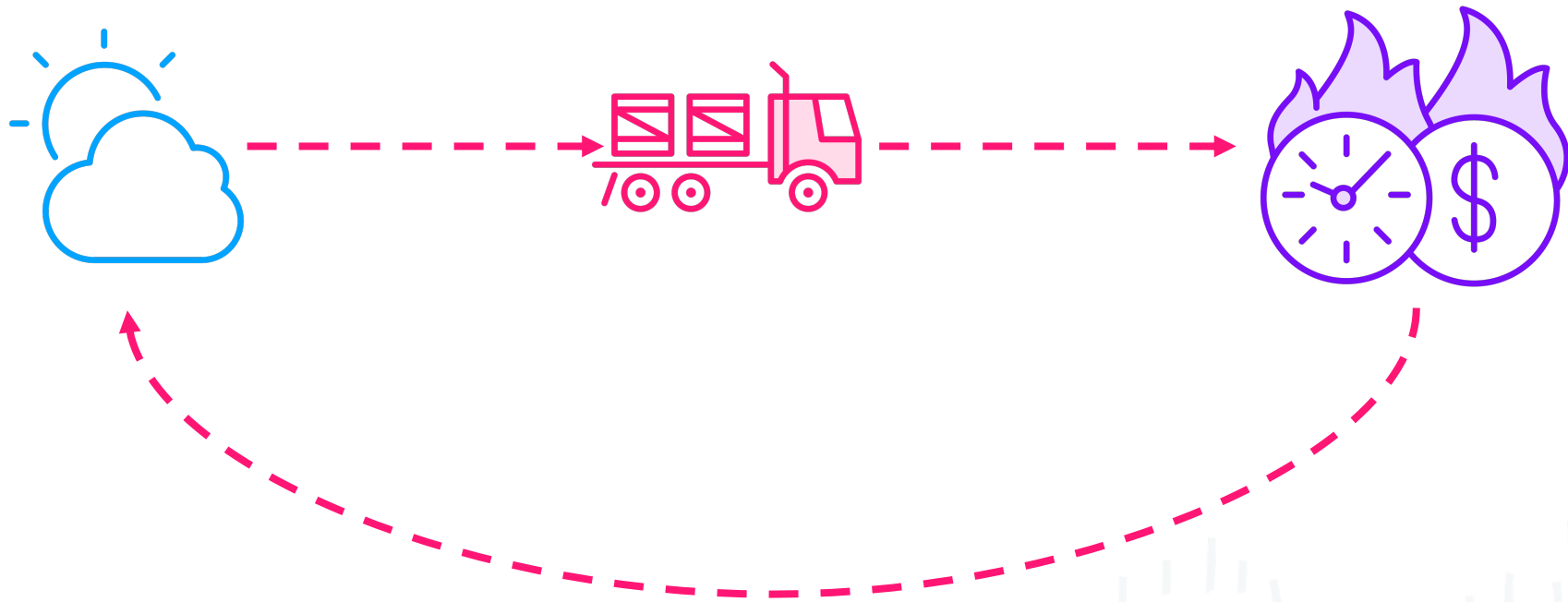


Effects

Inventory mismanagement

**Production and transportation
disruption**







Building and Deploying AI Agents



AutoGen

Open source

Developed by Microsoft

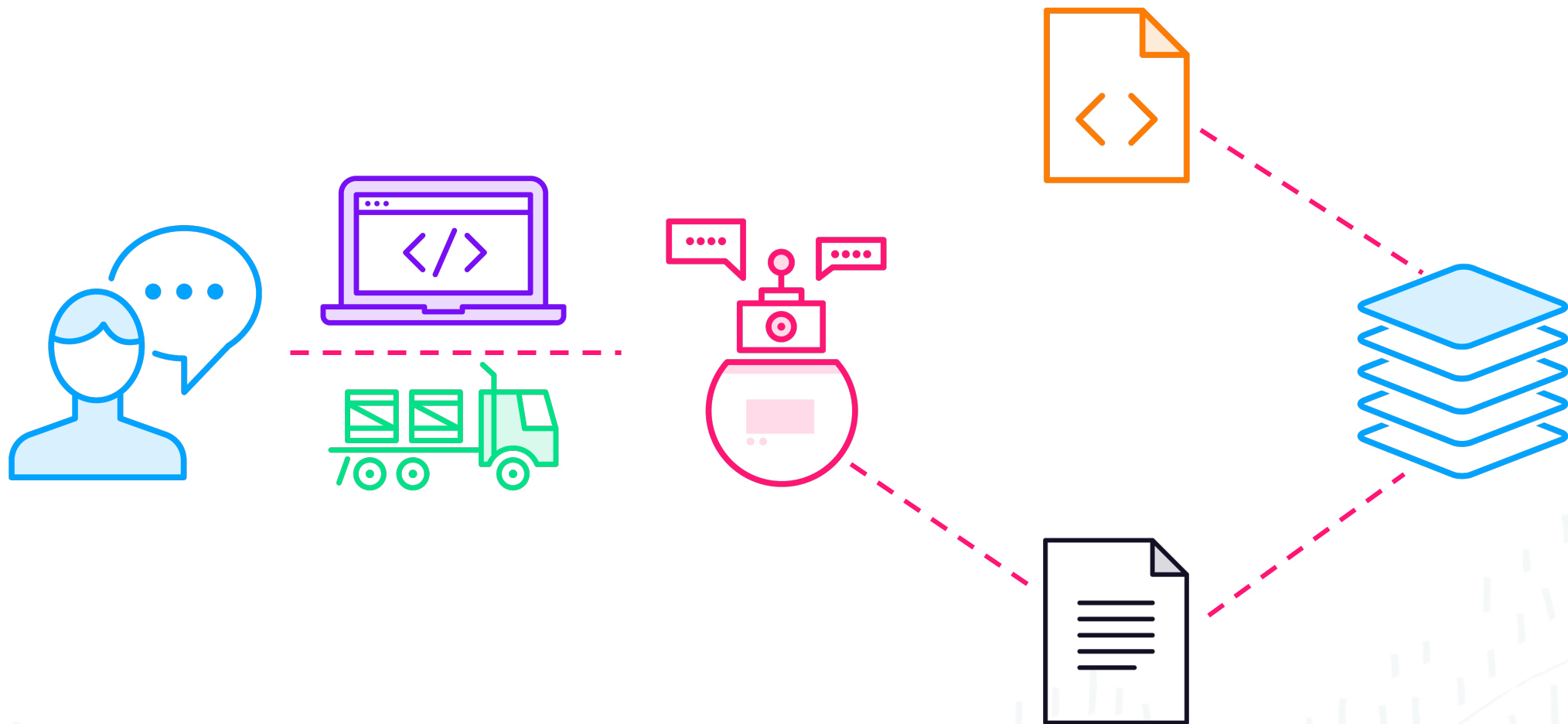
Advantages:

- Modular architecture
- Has built-in tools for monitoring and debugging

Limitations:

- Struggles with complex queries





OpenAI Agents SDK

Allows users to build multi-agent workflows

Advantages:

- Low learning curve
- Provides monitoring tools

Limitations:

- Limited to OpenAI's models



Gemini CLI

Command-line based tool that can interact with files, APIs or use search



LangChain

Highly-customizable

Can be coupled with any model



Agents Deployment

Closed source

Models are used through API

No GPU is needed

Open source

A GPU machine is needed



Closed Source Models Costs



Users are charged by used tokens.

GPT 3.5 charges 3 dollars for ~3000 characters.

Infrastructure costs are low.



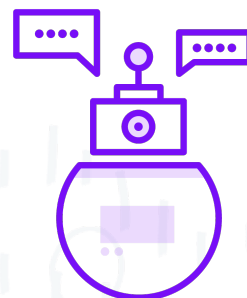
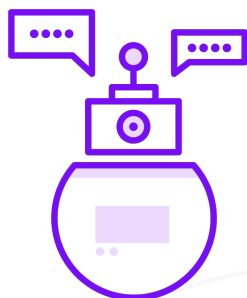
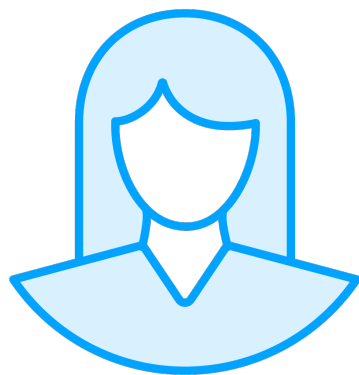
Open Source Models Costs

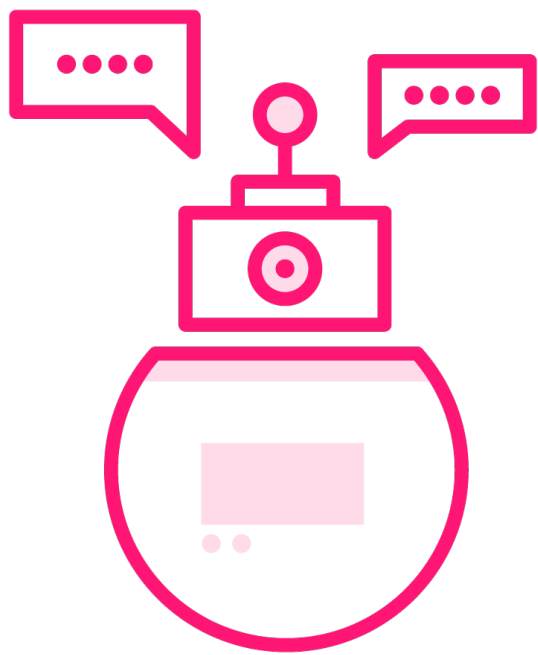


Requests are free.

Infrastructure costs can be high.





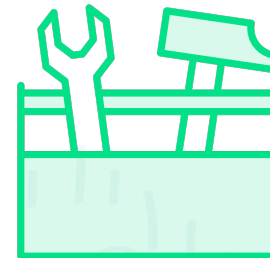
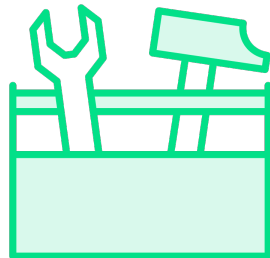
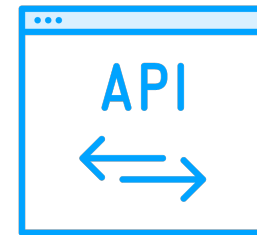


LangChain

OpenAI LLM

Python

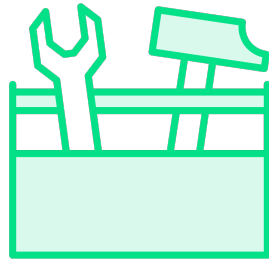
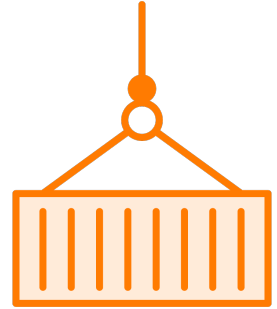
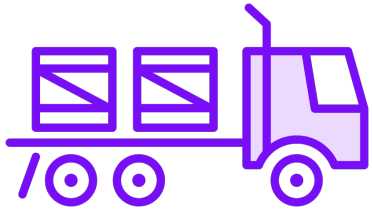






Predict Supply-chain Disruptions







Autonomous Freight Booking

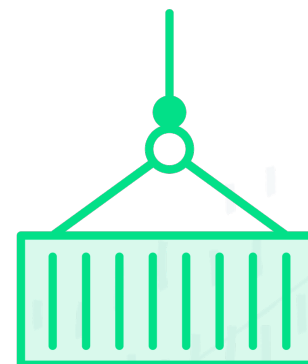
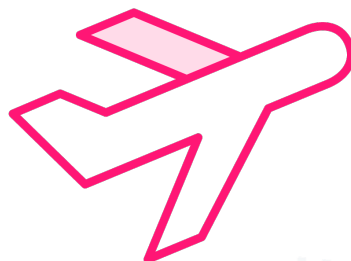
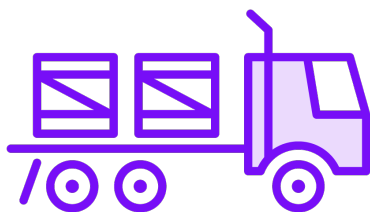




Every shipment is manually quoted

Operators sometimes forget to bid on spot freight during delays







Warehouse Automation



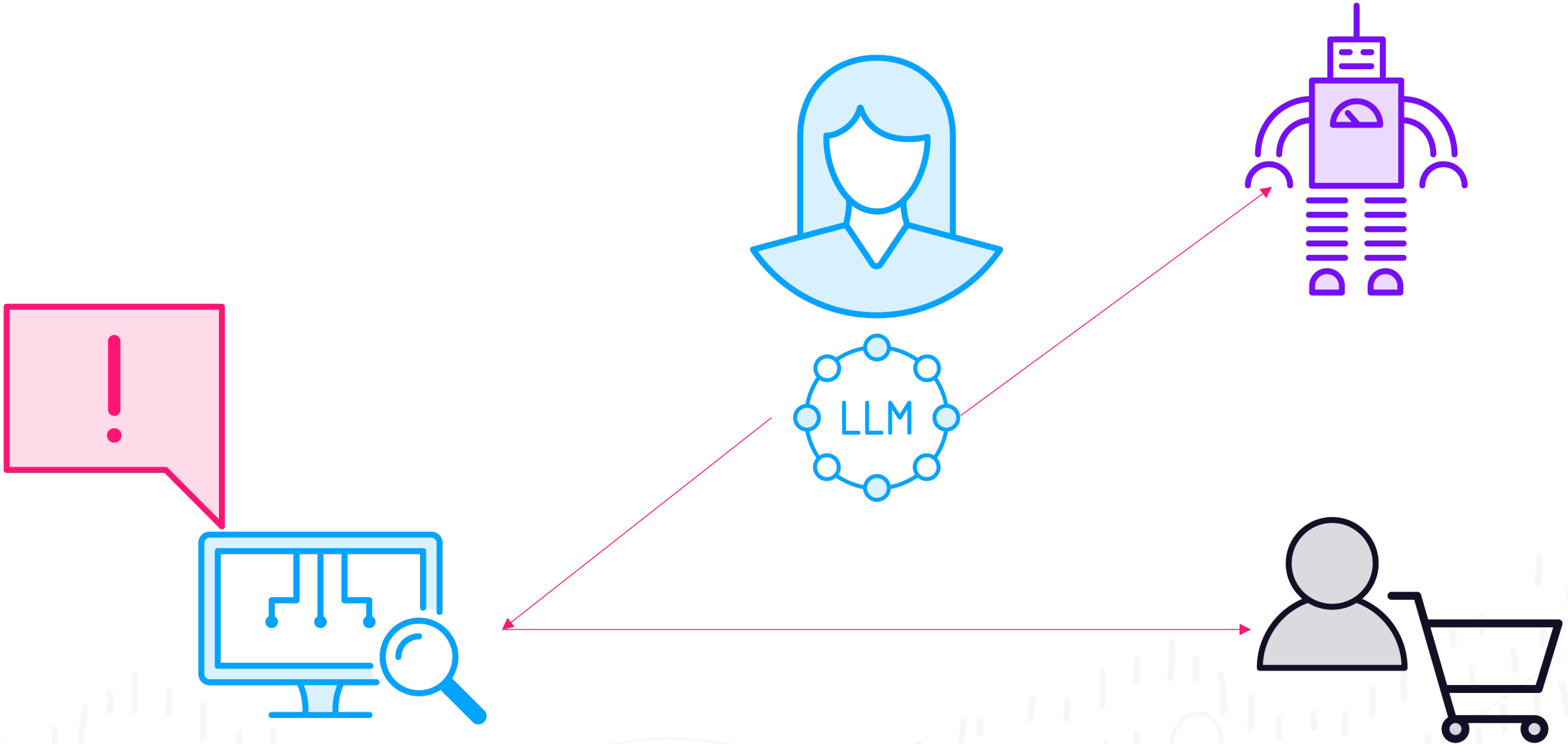


Inventory is checked manually

Orders are placed manually

SAP is updated

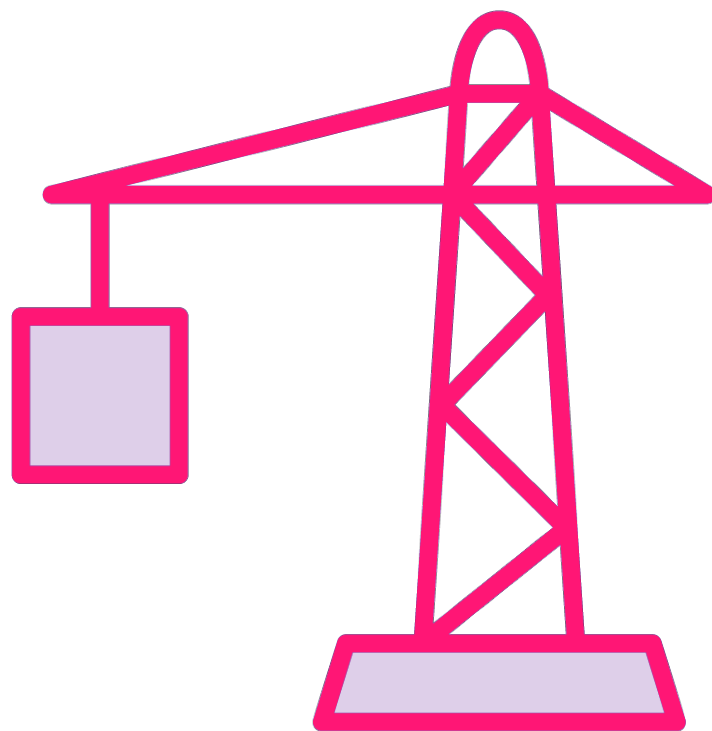


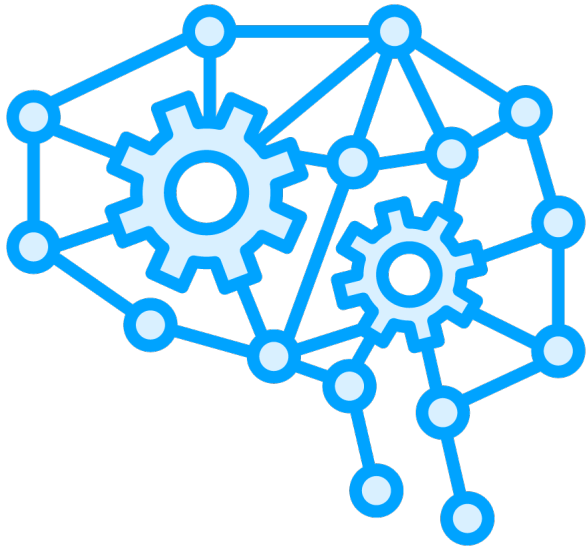




Other Use Cases



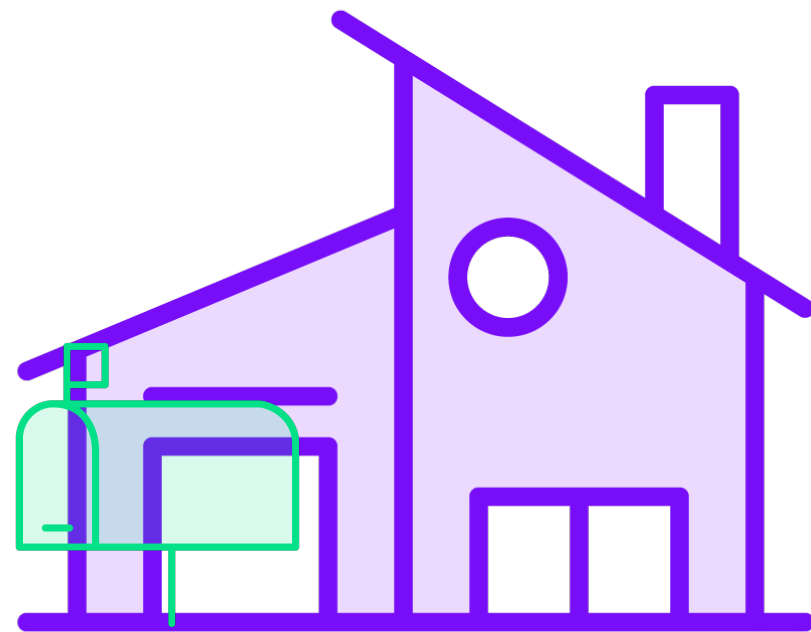
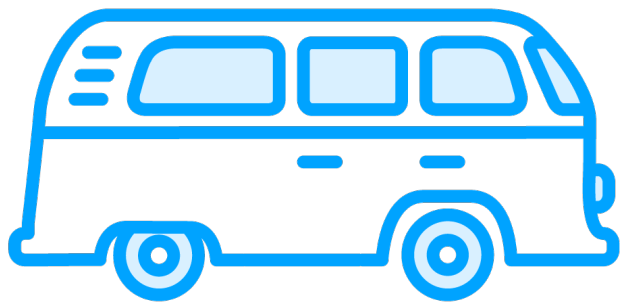




AI agents can:

- Read telemetry data
- Detect early warning signs of mechanical failure
- Schedule service automatically

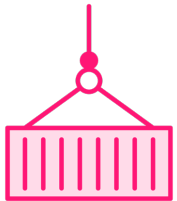




AI Agents Capabilities



Real-time customs clearance



Port scheduling



Supplier risk analysis



Agents free up time from manual tasks and enables logistics professionals to focus on strategy and resilience.



Summary

Optimize routes in real-time

Predict and respond to disruptions

Book freight

Automate warehouse tasks

Coordinate last-mile logistics



Thank you!

@bitheap_tech | www.bitheap.ch

